Lukas Schäfer

lukaschaefer.com | Google Scholar | lukas.schaefer@microsoft.com

Research Profile

I am a postdoctoral researcher at Microsoft Research, focusing on efficient learning algorithms for decision-making and multi-agent systems. I co-authored the first textbook on multi-agent reinforcement learning, and my research has been published at leading venues such as NeurIPS, ICML, TMLR, and AAMAS.

Work Experience

Postdoctoral Researcher Oct 2024 - Present
Microsoft Research *Cambridge, UK*

Researched **data-efficient imitation learning algorithms** for complex environments [2, 3, 15], co-developed the **WHAMM world model** [16] for real-time environment simulation deployed on Copilot Labs, and advised product teams on deploying foundation models.
Advisors: Sergio Valcarcel Macua and Katja Hofmann

Research Intern Apr 2023 - Oct 2023
Microsoft Research *Cambridge, UK*

Empirical study on visual encoders, including pre-trained foundation models, for data-efficient imitation learning in complex environments [13]. **Granted funding for an extension from 3 to 6 months** following a demo to stakeholders.
Advisors: Sam Devlin and Tabish Rashid

Research Intern Jul 2022 - Dec 2022
Huawei Noah's Ark Lab *London, UK*

Researched ensemble value functions to guide exploration and stabilise training in multi-agent reinforcement learning; accepted as **oral paper at AAMAS 2025** [4].
Advisor: David Mguni

Research Intern Nov 2020 - Mar 2021
Dematic - Technology and Innovation *Remote*

Designed high-performance **multi-agent warehouse simulator** and scalable learning algorithm, leading to **fellowship funding**, and a publication at **IROS 2024** [6].
Advisors: Aleksandar Krnjaic and Stefano V. Albrecht

Education

PhD, Data Science & Artificial Intelligence — **University of Edinburgh** — Dec 2019 - Oct 2024
Thesis: Efficient Exploration in Single-Agent and Multi-Agent Deep Reinforcement Learning
Advisors: Stefano V. Albrecht and Amos Storkey | **Award:** Passed with no corrections

MSc, Informatics — **University of Edinburgh** — Sep 2018 - Aug 2019
Thesis: Curiosity in Multi-Agent Reinforcement Learning
Advisor: Stefano V. Albrecht | **Award:** Distinction

BSc, Computer Science — **Saarland University** — Oct 2015 - Sep 2018
Thesis: Domain-Dependent Policy Learning using Neural Networks in Classical Planning
Advisor: Jörg Hoffmann | **Award:** 1.2 (within top 5%)

Publications

Citations on Google Scholar: 1,600+

Textbook

[1] S. V. Albrecht, F. Christianos, and **L. Schäfer**. *Multi-Agent Reinforcement Learning: Foundations and Modern Approaches*. MIT press, 2024. **50,000+ downloads**.

Journal and Conference Papers (Peer-Reviewed)

[2] **L. Schäfer**, A. Lemkhenter, P. Choudhury, C. Lovett, S. Nath, L. França, M. R. F. de Mendonça, A. Lamb, R. Islam, S. Sen, J. Langford, K. Hofmann, and S. V. Macua. "When does Predictive Inverse Dynamics Outperform Behavior Cloning?" In: *ICML*. 2026.

[3] S. Nath, A. Lemkhenter, P. Choudhury, C. Lovett, S. V. Macua, and **L. Schäfer**. "Augmentations for Robust and Efficient Imitation Learning in Streamed Video Games". In: *Conference on Games (CoG)*. 2026.

[4] **L. Schäfer**, O. Slumbers, S. McAleer, Y. Du, S. V. Albrecht, and D. Mguni. "Ensemble Value Functions for Efficient Exploration in Multi-Agent Reinforcement Learning". In: *AAMAS*. 2025.

[5] A. A. Fernandez, **L. Schäfer**, E. Villar-Rodriguez, S. V. Albrecht, and J. D. Ser. "Using Offline Data to Speed-up Reinforcement Learning in Procedurally Generated Environments". In: *Neurocomputing* (2024).

[6] A. Krnjaic, R. D. Steleac, J. D. Thomas, G. Papoudakis, **L. Schäfer**, A. W. K. To, K.-H. Lao, M. Cubuktepe, M. Haley, P. Börsting, and S. V. Albrecht. "Scalable Multi-Agent Reinforcement Learning for Warehouse Logistics with Robotic and Human Co-Workers". In: *IROS*. 2024.

[7] T. McInroe, **L. Schäfer**, and S. V. Albrecht. "Learning representations for control with hierarchical forward models". In: *TMLR* (2023).

[8] **L. Schäfer**, F. Christianos, J. P. Hanna, and S. V. Albrecht. "Decoupled Reinforcement Learning to Stabilise Intrinsically-Motivated Exploration". In: *AAMAS*. 2022.

[9] **L. Schäfer**. "Task Generalisation in Multi-Agent Reinforcement Learning". In: *AAMAS, DC*. 2022.

[10] R. Zhong, D. Zhang, **L. Schäfer**, S. V. Albrecht, and J. P. Hanna. "Robust On-Policy Data Collection for Data Efficient Policy Evaluation". In: *NeurIPS*. 2022.

[11] G. Papoudakis, F. Christianos, **L. Schäfer**, and S. V. Albrecht. "Benchmarking Multi-Agent Deep Reinforcement Learning Algorithms in Cooperative Tasks". In: *NeurIPS, Datasets and Benchmarks*. 2021.

[12] F. Christianos, **L. Schäfer**, and S. V. Albrecht. "Shared Experience Actor-Critic for Multi-Agent Reinforcement Learning". In: *NeurIPS*. 2020.

Workshops (Peer-Reviewed)

[13] **L. Schäfer**, L. Jones, A. Kanervisto, Y. Cao, T. Rashid, R. Georgescu, D. Bignell, S. Sen, A. T. Gavito, and S. Devlin. "Visual Encoders for Data-Efficient Imitation Learning in Modern Video Games". In: *ALA Workshop at AAMAS*. 2025.

[14] **L. Schäfer**, F. Christianos, A. Storkey, and S. V. Albrecht. "Learning Task Embeddings for Teamwork Adaptation in Multi-Agent Reinforcement Learning". In: *GenPlan Workshop at NeurIPS*. 2023.

Technical Blog Posts

[15] P. Choudhury, **L. Schäfer**, C. Lovett, K. Hofmann, and S. V. Macua. *Rethinking imitation learning with Predictive Inverse Dynamics Models*. Microsoft Research Blog. 2026.

[16] T. Rashid, V. Fragoso, C. Ke, Y. Cao, D. Bignell, S. Tan, **L. Schäfer**, S. Parisot, A. Lemkhenter, C. Lovett, P. Choudhury, R. Stevenson, S. V. Macua, A. Donnelly, D. Kennett, A. T. Gavito, L. Wen, J. Entenmann, K. Hofmann, and H. Zhang. *WHAMM! Real-time world modelling of interactive environments*. Microsoft Research Blog. 2025.

Teaching Experience

Textbook Author Mar 2022 - Dec 2024

Co-authored the **textbook on multi-agent reinforcement learning** [1] as an **equal contributor**, with responsibilities including curriculum design, authoring multiple chapters from start to finish, and co-developing the accompanying codebase and exercises. **Downloaded over 50,000 times** since pre-release in April 2023. Translated into Chinese.

Teaching Assistant Oct 2019 - Jun 2022
University of Edinburgh *Edinburgh, UK*

TA for the Reinforcement Learning course for three consecutive years; delivered lectures, designed coursework, and marked exams for 100+ students each year.

Voluntary Lecturer and Coach Sep 2017 - Oct 2017
Saarland University *Saarbrücken, Germany*

Delivered daily lectures on formal languages and logic to 250 participants in the orientation course for incoming computer science students; received the **BESTE-award** for special student commitment (Saarland University, 2017).

Teaching Assistant Oct 2016 - Mar 2017
Saarland University *Saarbrücken, Germany*

Taught weekly tutorials on functional programming, and complexity theory to undergraduates; co-designed teaching material and marked weekly tests and exams.

Supervision and Mentorship Experience

Consultation & Mentorship Sep 2025 - Present
Microsoft Research *Cambridge, UK*

Consulting team of 15 engineers applying vision-language models for production systems on machine learning practices; closely mentoring a graduate-level software engineer.

Supervision of PhD Research Interns Apr 2025 - Present
Microsoft Research *Cambridge, UK*

Supervising PhD research intern on video models for robot planning. Supervised Somjit Nath (McGill, Canada) on data augmentations for efficient imitation learning; resulted in **Conference on Games** publication [3], and informed research and product teams.

Supervision of Visiting PhD Student May 2022 - May 2023
University of Edinburgh *Edinburgh, UK*

Supervised visiting PhD student Alain Andres Fernandez (Tecnalia, Spain) on online adaptation of offline pre-trained AI agents during a 3-month visit and subsequent remote collaboration; resulted in a **Neurocomputing Journal** publication [5].

Supervision of Master's Students Sep 2021 - May 2024
University of Edinburgh *Edinburgh, UK*

Supervised three Master's students during their MSc dissertation projects:

- Rahat Santosh: Integrating Agent Modelling in Multi-Agent Reinforcement Learning Algorithms
- Rujie Zhong: Data Collection for Policy Evaluation in Reinforcement Learning
  Resulted in main conference publication at **NeurIPS 2022** [10]
- Panagiotis Kyriakou: Reinforcement Learning with Function Approximation in Continuing Tasks: Discounted Return or Average Reward?

Funding

Awarded Funding

Dec 2019 - Jun 2024	£58,731	**Principal's Career Development Scholarship** PhD scholarship from University of Edinburgh
Apr 2023 - Oct 2023	£12,860	**Microsoft Research internship extension** funding to extend research internship from 3 to 6 months; granted after demo of project progress
Sep 2018 - Aug 2019	16,500 €	**DAAD graduate scholarship** postgraduate scholarship from German Academic Exchange Service
Sep 2018 - Aug 2019	£300	**Stevenson Exchange Scholarship** postgraduate scholarship to support studies in Scotland

Contributions to Funding Awards

Royal Academy of Engineering, Industrial Fellowship 2022

My internship project at Dematic initiated and laid the foundation for a multi-year research collaboration funded with **£250,000** through a **Industrial Fellowship of the Royal Academy of Engineering** between Dematic and the University of Edinburgh on scalable multi-agent reinforcement learning for warehouse logistics. The fellowship was held by my primary PhD supervisor, Prof. Stefano V. Albrecht.

Awards

Young Researcher Attendee Sep 2022
Heidelberg Laureate Forum *Heidelberg, Germany*

Selected as one of 100 international young researchers in computer science to participate in the prestigious Heidelberg Laureate Forum where I had the opportunity to network and discuss research with laureates in mathematics and computer science.

Open-Source Software Contributions

EPyMARL – Core Contributor

Core contributor to EPyMARL codebase for multi-agent reinforcement learning research, developed as part of a benchmarking paper [11]. **Over 700 stars on Github**.

MARL Textbook Codebase – Core Contributor

Core contributor to the multi-agent reinforcement learning textbook [1] codebase designed for use in teaching. **Over 600 stars on Github**.

Skills

Natural Languages	German (native), English (fluent), French (basic), Chinese (basic)
Programming	Python (primary), C++, C, Bash, Java

Research Community Engagement

Reviewing

- 3-time **best reviewer award** for ICML conference (2022, 2024, 2025)
- Journals: Transactions on Machine Learning Research (TMLR, 2024)
- Conferences: NeurIPS (2021, 2022, 2023), ICLR (2026), ICML (2021, 2022, 2023, 2024, 2025, 2026), AAMAS (2022, 2023, 2024), Conference on Games (CoG, 2026), RLDM (2025), RLC (2024)
- Workshops: NeurIPS Pre-registration experiment workshop (2020)

Organisation

Lead Organiser Jul 2024 - Mar 2025
UK Multi-Agent Systems Symposium 2025 *London, UK*

Co-lead organiser of the UK Multi-Agent Systems Symposium 2025 with 200 participants in collaboration with the Alan Turing Institute and King's College London.

Lead Organiser and Host Sep 2020 - Sep 2022
RL Reading Group, University of Edinburgh *Edinburgh, UK*

Organised and hosted RL reading group at University of Edinburgh with speakers from leading industry and academic research groups, including Google DeepMind, FAIR, Oxford University, McGill University, and National University of Singapore.

Selected Invited Talks

Mar 2026	**University of Edinburgh**, RL & Agents Reading Group Understanding Predictive Inverse Dynamics Models for Offline Imitation Learning
Feb 2026	**Imperial College**, I-X Seminar Decision-Making in Modern Video Games: From Game-Playing Agents to World Models
Jan 2026	**Cohere Labs Seminar** Decision Making in Modern Video Games: From Human Play to World Models
Dec 2025	**ELLIS UnConference, Interactive Learning and Interventional Representations (ILIR) Workshop** Exploiting State and Action Uncertainty for Imitation Learning using Inverse Dynamics Models
Oct 2025	**University of Sheffield**, ML Seminar Series Decision Making in Modern Video Games: From Human Play to World Models
Jul 2025	**Belgium-Netherlands Workshop on Reinforcement Learning (BeNeRL)** Decision Making in Video Games
Nov 2024	**Gazi University Turkey**, AI Research & Big Data Seminars An Introduction to the Multi-Agent Reinforcement Learning Textbook
May 2024	**Microsoft Research Cambridge** Efficient and Scalable Decision Making In Complex Environments
Mar 2024	**University of Maryland**, MARL Reading Group An Introduction to MARL Textbook and EPyMARL Codebase
Feb 2024	**Stanford University**, Stanford Intelligent Systems Laboratory Sample-Efficient Multi-Agent Reinforcement Learning
Jul 2022	**University of California, Berkeley**, RL Reading Group Deep Reinforcement Learning for Multi-Agent Interaction